

Introduction to Machine Learning Safety

Exercise Sheet 6: Explainability

May 29, 2026
Questions & Answers

Explainability Concepts

Exercise 6.1: Why Explainability? (optional)

What are advantages of explainable models in a safety-critical context? What are potential disadvantages or limitations of current explainability methods?

Answer:

Advantages in safety-critical contexts:

- **Trust and verification:** Engineers and regulators can inspect why a model made a decision rather than blindly trusting its output. In autonomous driving, medical diagnosis, or aviation, this is legally and ethically necessary.
- **Bias and shortcut detection:** Explainability can reveal if a model learned spurious correlations (e.g., predicting "pedestrian present" based on sky color), enabling early correction before deployment.
- **Debugging and improvement:** When a model fails, explanations help diagnose where the failure originates - wrong features, distribution shift, or underrepresented training data.
- **Accountability:** In regulated domains (e.g., EU AI Act), decisions affecting humans must be explainable. Models that can provide justifications satisfy compliance requirements.
- **Human-AI collaboration:** Operators can override or correct AI decisions more effectively when they understand the model's reasoning.

Disadvantages and limitations:

- **Faithfulness problem:** Explanations may not actually reflect the model's internal computation. A saliency map shows input sensitivity, not necessarily the causal reasoning chain.
- **Approximation errors:** Methods like LIME approximate a complex model locally with a simpler one - this approximation can be inaccurate or misleading.
- **Human confirmation bias:** Users tend to trust explanations that look reasonable even if they are wrong. A plausible-looking heatmap can create false confidence.
- **Computational cost:** Methods like occlusion or SHAP can be extremely expensive on large models or high-dimensional inputs.
- **No ground truth:** There is no universal way to objectively verify whether an explanation is correct, making evaluation inherently difficult.

Exercise 6.2: Local vs. Global Explainability

Describe the difference between local and global explainability. Name one method for each kind and explain what question it answers.

Answer:

Dimension	Local	Global
Scope	Explains a single prediction	Explains the model's overall behavior
Question answered	"Why did the model predict X for this input?"	"How does the model generally make decisions?"
Example method	LIME	Permutation Feature Importance

Local - LIME:

LIME perturbs a single input (e.g., masks patches of an image), observes how the model output changes, and fits a simple interpretable model (e.g., linear) locally around that point. It answers: "Which parts of this specific image most influenced this specific prediction?"

Global - Permutation Feature Importance:

Randomly shuffles one feature across the entire dataset and measures the average drop in model performance. It answers: "Which features does the model rely on most, across all predictions?"

Exercise 6.3: Saliency vs. Occlusion

Briefly describe the saliency method and the occlusion method. What is one advantage and one disadvantage of each compared to the other?

Answer:

Saliency Maps:

Compute the gradient of the output with respect to each input pixel. A large gradient magnitude means that small changes to that pixel strongly affect the prediction. The map is produced in a single forward + backward pass.

- **Advantage:** Computationally very cheap - only one forward and one backward pass required, regardless of input size.
- **Disadvantage:** Gradients can be noisy and sensitive to local pixel perturbations ("gradient saturation"), often highlighting irrelevant regions. They reflect local sensitivity, not necessarily causal importance.

Occlusion:

Systematically slide a patch/mask over the input, replace it with a neutral value (e.g., gray, zero), and record how much the model's confidence drops at each position. The resulting heatmap shows which regions, when hidden, most reduce the prediction score.

- **Advantage:** Intuitively interpretable and faithful in a causal sense - it directly measures the effect of removing information, which is a more meaningful notion of importance.
- **Disadvantage:** Computationally expensive - requires a separate forward pass for every patch position, scaling poorly with image resolution and patch size.

Exercise 6.4: Chain-of-Thought Fidelity

Part 1: What does it mean for a thinking trace to be faithful? Why is faithfulness hard to verify?

Answer:

A thinking trace is faithful if it accurately reflects the actual internal computational process the model used to arrive at its answer - i.e., the steps described in the trace are the real causal steps, not a post-hoc rationalization.

Faithfulness is hard to verify because:

- We have no direct access to the model's internal reasoning - only its weights and activations, which do not map neatly onto human-readable steps.
- The model generates the trace using the same token-prediction process as the answer itself, so the trace may simply be a plausible-sounding narrative rather than an account of what computations actually occurred.
- There is no ground truth "true reasoning path" to compare against.

Part 2: What is simulatability as a measure of thinking-trace quality? Give an example scenario where simulatability is satisfied but the trace is still unfaithful.

Answer:

Simulatability means that a human (or another system) can read the thinking trace and, by following its steps, arrive at the same answer the model produced - i.e., the trace is internally consistent and human-followable.

Counterexample: A model is asked whether a loan application should be approved. The trace writes: "The applicant has a high credit score and stable income, therefore approve." A human following this trace would also approve - so it is simulatable. However, the model actually classified based primarily on the applicant's ZIP code (a proxy for demographic data) which correlates with approval in the training data. The stated reasoning is consistent but does not reflect what actually drove the decision.

Part 3: What does counterfactual simulatability add? Why is it a stricter test?

Answer:

Counterfactual simulatability asks: if a key fact in the trace were changed, would the model's answer change in the way the trace predicts?

Example: if the trace says "the credit score is high, therefore approve", then artificially lowering the credit score input should cause the model to predict "reject". If it still approves (because it was actually using ZIP code), the trace fails the counterfactual test.

This is stricter than plain simulatability because it tests causal consistency: the stated reasons must actually be the driving factors, not just narrative post-hoc justifications that happen to be plausible.

Part 4: Why does an unfaithful thinking trace pose a safety risk? Give one concrete example.

Answer:

An unfaithful thinking trace poses safety risks because it creates false transparency - operators believe they understand and can audit the model's decisions, but the stated reasoning does not correspond to reality.

Concrete example: An autonomous vehicle's perception system produces a chain-of-thought: "I see a pedestrian at the crosswalk; I detect motion; I am braking." A safety engineer reviews this trace and concludes the system is reasoning correctly. In reality, the model brakes because it detected a certain road texture pattern correlated with pedestrians in training data. In a new city with different road surfaces, this spurious correlation fails, the model does not brake, and the unfaithful trace gave the engineer no warning - because the trace always looked correct even as the underlying computation was unreliable.

Practical: Explaining the CARLA Model

Exercise 6.5: Applying an Explainability Method

Choose one local explainability method from the lecture: occlusion, CAM, Grad-CAM, or saliency maps. Apply it to each of your three CARLA models.

Answer:

Part 1: Method description and motivation

(To be completed with code implementation)

CODE SECTION - Grad-CAM / chosen method implementation pending

Part 2: Visualize explanations for 5 correctly classified images

(To be completed with code implementation)

CODE SECTION - Heatmap overlays on correctly classified images pending

Part 3: Do highlighted regions correspond to relevant objects?

(To be completed after visual inspection of heatmaps)

CODE SECTION - Analysis of highlighted regions pending

Part 4: Select 3 misclassified images and analyse explanations

(To be completed with code implementation)

CODE SECTION - Misclassification explanation analysis pending

Exercise 6.6: Explainability as a Diagnostic Tool

Part 1: Suppose your explanation method reveals that your model predicts "pedestrian present" based primarily on a region of the sky rather than the pedestrian itself. What would this imply about the model's generalization? What could cause this behavior?

Answer:

This would imply that the model has learned a spurious correlation rather than a genuine causal feature - i.e., the model is not actually detecting pedestrians, but instead detecting a background feature that happened to co-occur with pedestrians in the training data.

Implications for generalization:

- The model would fail as soon as the sky appearance changes - e.g., overcast vs. sunny days, different times of day, or different geographic locations with different sky colors.
- The model would not generalize to scenes where the spurious correlation no longer holds, even if the relevant object (pedestrian) is clearly visible.
- This is an example of shortcut learning or a Clever Hans failure - the model achieves high training accuracy via a wrong mechanism.

Possible causes:

- Dataset bias: In the training set, pedestrian images may systematically come from a specific time of day (e.g., daytime with blue sky) while non-pedestrian images were captured under different sky conditions.
- Lack of diversity: If the CARLA simulator always rendered pedestrian scenes with clear sky and non-pedestrian scenes with a different background, the model exploited this confound.
- Insufficient data augmentation: Without augmentation that varies sky appearance independently of pedestrian presence, the model cannot disentangle the two signals.

Part 2: Test your models on images from different conditions (different town layout, nighttime, foggy weather). Apply the same explanation method.

(a) Do the highlighted regions still correspond to the relevant objects?

Answer:

(To be completed with code implementation and OOD testing)

CODE SECTION - OOD condition testing pending

(b) Do you observe evidence that the model relied on background or spurious features specific to the original training conditions?

Answer:

(To be completed after visual inspection across conditions)

CODE SECTION - Spurious feature analysis across conditions pending

(c) How do the model's accuracy and explanation quality change across conditions?

Answer:

(To be completed with quantitative accuracy measurements and explanation comparison)

CODE SECTION - Accuracy and explanation quality analysis pending